CALCIUM GLUCONATE

Hyperkalaemia: protects heart

Mg^{2+} toxicity antidote

CHEMICAL NAME Calcium D-gluconate ($C_{12}H_{22}CaO_{14}\cdot H_2O$); 10% solution = 0.22 mmol (8.9 mg) Ca^{2+} per mL (10 mL = 2.2 mmol)

CLASS Calcium salt (electrolyte replacement; cardioprotectant; antidote)

TRADE NAMES Calcium gluconate 10% injection · Calcium-Sandoz · Calcium gluconate tablets

THERAPEUTIC USES / INDICATIONS

- Hyperkalaemia with ECG changes (cardiac membrane stabilisation; first step).
- Severe/symptomatic hypocalcaemia (tetany, seizures, laryngospasm, prolonged QT).
- Magnesium sulfate toxicity (antidote).
- Calcium-channel blocker overdose; hydrofluoric acid burns (gel); massive transfusion (citrate).

ADMINISTRATION

- **IV** slowly (10 mL 10% over 5–10 min) with cardiac monitoring; or infusion.
- Large vein — less irritant than calcium chloride but still causes necrosis if extravasated.
- Do not mix with sodium bicarbonate or phosphate (precipitate); flush line.
- Never IM or SC (necrosis).

DOSAGES

- **Hyperkalaemia:** 30 mL of 10% IV over 5–10 min; repeat if ECG changes persist (effect lasts 30–60 min).
- **Acute hypocalcaemia:** 10–20 mL of 10% IV over 10 min, then infusion: 100 mL of 10% in 1 L saline or 5% glucose at 50–100 mL/h, adjusted to calcium.
- **Mg^{2+} toxicity:** 10 mL of 10% IV over 10 min.
- **Children:** 0.11 mmol/kg (0.5 mL/kg of 10%, max 20 mL).

PHARMACOKINETICS

- **Onset:** IV immediate. **Duration:** membrane effect 30–60 min.
- Calcium distributes to ECF, bone; excreted in urine and faeces.

PHARMACODYNAMICS

- Raises **extracellular calcium** → restores threshold potential in myocardium (antagonises hyperkalaemia membrane effects without lowering K^+).
- Corrects hypocalcaemia; antagonises Mg^{2+} neuromuscular block.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Tingling, sensation of warmth, chalky taste.
- Flushing.
- Injection-site irritation.

ADVERSE EFFECTS (SERIOUS)

- Bradycardia, arrhythmias, cardiac arrest with rapid injection or in patients on digoxin (digoxin toxicity).
- Hypercalcaemia.
- Tissue necrosis/calcinosis with extravasation.

CONTRAINDICATIONS & CAUTIONS

- Hypercalcaemia; ventricular fibrillation (not in cardiac arrest unless hyperkalaemia/hypocalcaemia).
- **Caution:** digoxin therapy (give slowly with monitoring), renal stones, sarcoidosis.

KEY INTERACTIONS

- **Digoxin:** arrhythmias — slow administration, monitor.
- **Sodium bicarbonate, phosphate, ceftriaxone:** precipitation.
- **Thiazides:** hypercalcaemia (chronic).

NURSING MANAGEMENT

- Continuous ECG monitoring; give slowly.
- Check line patency; large vein.
- Check serum calcium (adjusted for albumin or ionised), K^+, Mg^{2+}.
- Remember calcium does not lower K^+; insulin/glucose and other measures needed.
- Observe for tetany signs (Chvostek, Trousseau) in hypocalcaemia.

PATIENT EDUCATION

- A warm feeling and chalky taste during injection are common.
- Report pain at the drip site or palpitations.

KEY POINTS

- IV calcium — cardioprotection in hyperkalaemia (10–30 mL 10%).
- Slow IV with ECG; caution with digoxin.
- Treats hypocalcaemia; antidote for Mg^{2+}.
- Never mix with bicarbonate.

SOURCES CHECKED **[966]** Calcium Gluconate 10% solution for injection/infusion BP (UK SmPC) **[967]** Turner et al., Endocr Connect 2016 **[448]** UKKA hyperkalaemia guideline (2023)

Full references p. 432–457